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Research paper

Challenges of sustainable groundwater management for large scale irrigation under changing climate in Lower Ganga River basin in India

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ABSTRACT

Groundwater based irrigation is one of the key factors for rapid agricultural development in India. Therefore, understanding the sustainability of irrigation water management is utmost important for agriculture. This study investigated the sustainability of groundwater based irrigation by identifying recent (1996–2017) trend in groundwater depth using Mann-Kendall (MK) test or Modified MK and Sen's slope analysis. This study also explored the relationships of groundwater depth with rainfall and irrigated areas in lower Ganga River basin in India with an area 195,601 km^2. Results showed that almost 81.78%, 66.22%, 74.38% and 66.79% wells for pre-monsoon, monsoon, post-monsoon and winter seasons respectively showed increasing trends in groundwater depth. The rapid increasing trend (slope: 8.95–104.46 cm/year) in monsoon season indicates shallow groundwater resource depletion and groundwater abstraction exceeds the groundwater recharge. Moreover, dry seasons increasing trends infer that shallow pumps will not functioning in future as a result irrigation cost will increase. Although irrigated areas remain almost same (1997–2014), decreasing rainfall along with abstraction of groundwater during dry season is primarily identified as main driving force for groundwater depletion. Further study is necessary to characterize the groundwater depletion. Planned water management is urgent for ensuring sustainable irrigation water management.

1. Introduction

Water is one of the crucial natural resources for agricultural, economic and environmental development (Rahman et al., 2017 and Rahman et al., 2020a,b). However, global groundwater resource has already affected by different types of problems such as rapid depletion (Park et al., 2011; Dalin et al., 2017), large scale pollution and seawater intrusion (Gorelick and Zheng, 2015). Global climate change has also affected the groundwater resources as well as hydrological cycle (Stagl et al., 2014). Groundwater dependent irrigation has significantly increased in developing countries such as India over the last half of the twenty century (Siebert et al., 2010). India is the highest groundwater user in the globe and almost 250 billion m^3 has withdrawn each year (Shah et al., 2007; Singh and Singh, 2002). Groundwater is the single most sources of water for irrigation and about 65% irrigated water comes from groundwater (Siebert et al., 2010) in India. Such huge withdrawal of groundwater by millions of independent farmers is called "Silent Revolution" (Llamas and Martínez-Santos, 2005). This withdrawal has largely occurred without any hydrological planning resulting high groundwater depletion in developing county like Bangladesh (Ali et al., 2012; Rahman et al., 2016). Irrigated agriculture causes groundwater depletion, however, food security and rural livelihood in India depends on irrigated agriculture. About 70% of the food grains come from irrigated agriculture (Asoka et al., 2017) in India. Therefore, sustainability of irrigation water management is utmost important for understanding food security as well as environmental and socio-economic development under the changing climate.

Although understanding the sustainability of water resources management is very important for developmental perspectives; there is no particular approach that can deal with the problem (Harmancioglu et al., 2013). There are several traditional approaches such as groundwater recharge, safe yield and groundwater footprint for understanding

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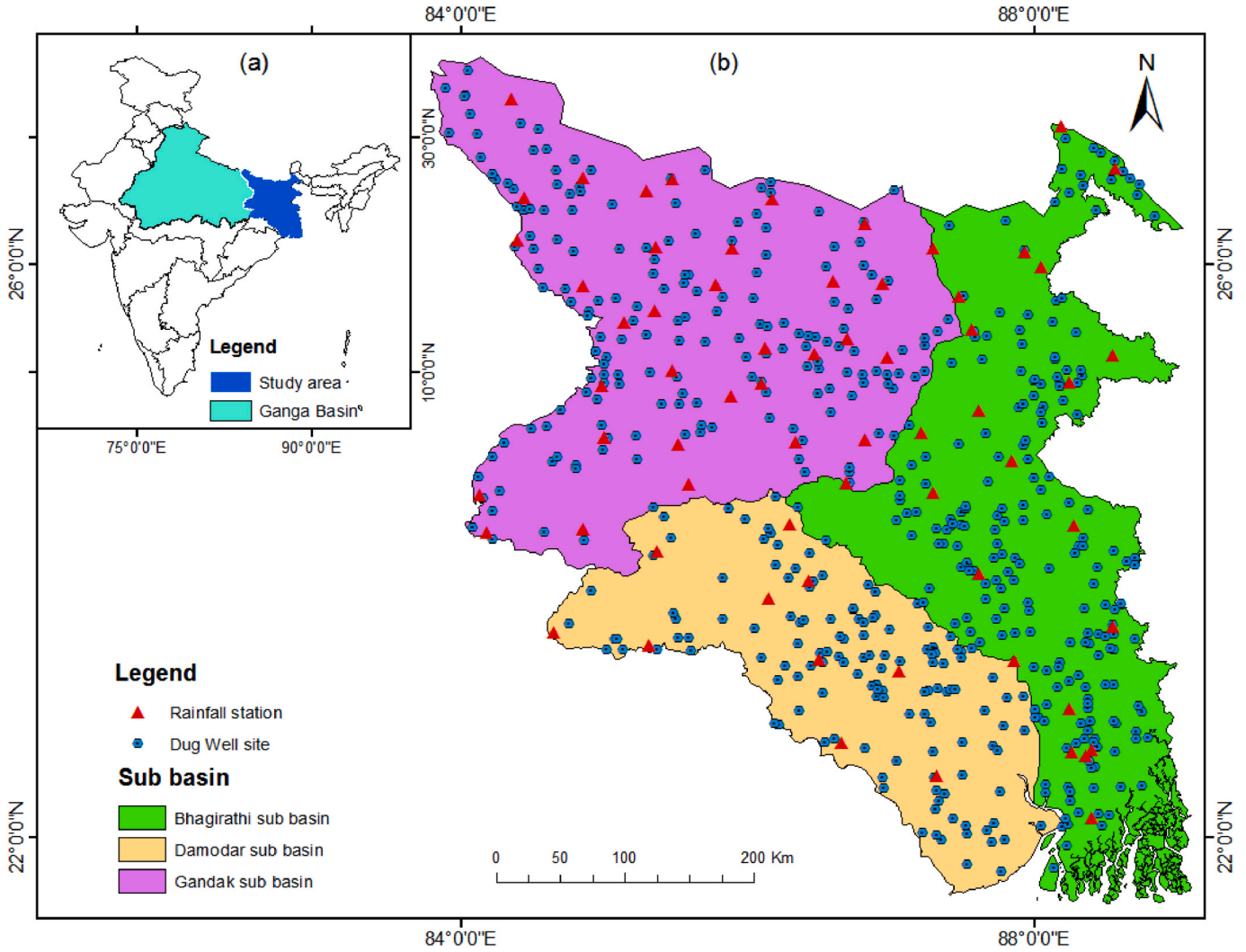


Fig. 1. Study area with location of groundwater depth monitoring dug wells and rain gauge measuring stations.

sustainability (Gleeson et al., 2012; Harmancioglu et al., 2013; Rahman et al., 2018a). However, these traditional approaches are not suitable for understanding sustainability especially under the changing climate (Gleeson et al., 2012; Balleau, 2013). Identifying the trend in groundwater storage changes derived from remote sensing observation data such as Gravity Recovery Climate Experiment (GRACE) has attracted attention for understanding global groundwater sustainability over the last few decades (Thomas et al., 2017). Studies (Rodell et al., 2009) also reported the declining trend in groundwater storage in India based on GRACE data; however, identification of trend in observation data is limited in India (Asoka et al., 2017). Furthermore, identifying trends in observation data (*In situ* observation) and characterizing the trends are necessary for preparing sustainable water resources management plans to cope with changing climate (Rahman et al., 2016, 2017b).

Therefore, the aim of this study was to develop a framework for assessing the sustainability of groundwater management for irrigation water management. To fulfil the goal of this study, we identified the trends in long-term recent groundwater data. Trend was characterized by spatio-temporal changes in rainfall and irrigated areas in Lower Ganga River Basin (LGRB) in India. To accomplish this study, we used non-parametric Mann-Kendall (MK) (Mann, 1945; Kendall, 1975) test or Modified MK (Hamed and Rao, 1998) based on the properties of the observation data. The slope of the changes was estimated by Sen's estimator (Sen, 1968). MK and MMK tests are the robust methods for identifying trends in hydro-meteorological variables and can be applied in missing data (Tabari et al., 2012; Nalley et al., 2013; Rahman et al., 2018b).

2. Study area and methods

2.1. Study area

The LGRB is located in the northeastern part of India, it comprises four sub-basins namely Bhagirathi, Damodar, Gandak and Sone, due to the unavailability of groundwater data Sone sub-basin excluded from our study area. The present study area extends from 21° 36′ 30″N to 27° 25′ 30″N latitudes and 83° 47′ 30″ E to 89° 05' E longitudes, having the geographical area of 195,601 km^2 (Fig. 1). The basin includes Bihar, Jharkhand and West Bengal states and covers 78 districts in Eastern India and home to approximately 227 million peoples. Despite the presence of numbers of large rivers (Bhagirathi, Damodar, Gandak and Sone), most of the cultivators of this region are dependent on groundwater for agriculture, especially for *Rabi* crop cultivations. *Boro* rice is highly irrigated crops, cultivated almost 52.14% (2014) of the total cultivated area during November to May. Furthermore, in recent years, Boro rice cultivation has been extensively increased in West Bengal and Bihar (Singh, 2002). Hence, groundwater resource is highly exploited in the basin. The study area divided into 9 principal aquifer systems. Alluvium aquifer of quaternary age occupied the maximum areas of the basin and covering the northern, southern and eastern parts of the LGRB.

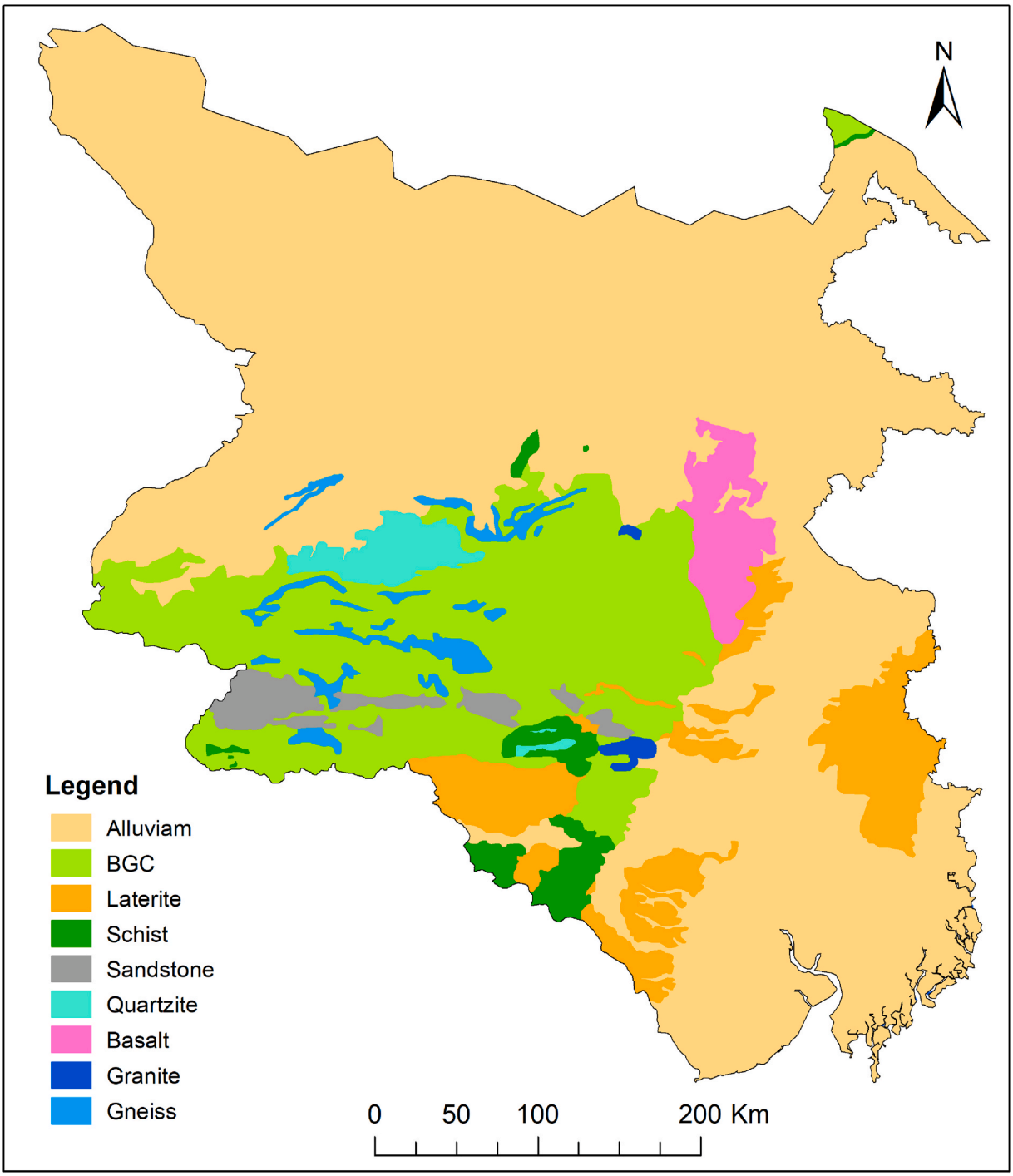


Fig. 2. Principal aquifer systems of Lower Ganga River Basin.

The depth of deep-water level of alluvium is 2–40 mbgl, whereas, thickness of weathered zone is 2–1200 m (CGWB, 2020). Likewise, laterite spread over some pockets of eastern and western parts, which has the moderate depth of groundwater level (5–20 mbgl). The rest of the area of western part of the basin are occupied by BGC, schist, sandstone, quartzite, basalt, granite and gneiss aquifer (Fig. 2).

2.2. Data

Recent seasonal groundwater observation data (1996–2017) of more than thousands dug wells located in the LGRB were collected from India-Water Resource Information System portal (http://cgwb.gov.in/GW-data-access.html). The locations of the analyzed dug wells are shown in Fig. 1. However, data of many dug wells have long missing records. This study discarded the dug wells with long missing records and analyzed the groundwater observation data of 527 dug wells for pre-monsoon (March to May), monsoon (June to September), post-monsoon (October to November) and winter (December to February) seasons. We also analyzed the irrigated areas in West Bengal, Bihar and Jharkhand States. Data were collected from data.gov portal in India (https://data.gov.in/). This study also analyzed the rainfall data for the period of 1980–2010. Rainfall data base was compiled from India water portal (https://www.indiawaterportal.org/met_data/).

Table 1
Number of observation wells with identified trends by MK & MMK Test in recent (1996–2017) groundwater depth at different confidence level in the study area.

Increasing Trends in Groundwater Depth	Pre monsoon		Monsoon		Post monsoon		Winter	
	Number of Observation Wells	% of Wells	Number of Observation Wells	% of Wells	Number of Observation Wells	% of Wells	Number of Observation Wells	% of Wells
99% Confidence Level	128	24.29	62	11.76	70	13.28	56	10.63
95% Confidence Level	78	14.80	55	10.44	62	11.76	39	7.40
90% Confidence Level	41	7.78	34	6.45	37	7.02	31	5.88
Insignificant	184	34.91	198	37.57	223	42.31	226	42.88
Total	**431**	81.78	349	66.22	392	74.38	352	66.79
Decreasing Trends in Groundwater Depth	**Number of Observation Wells**	**% of Wells**	**Number of Observation Wells**	**% of Wells**	**Number of Observation Wells**	**% of Wells**	**Number of Observation Wells**	**% of Wells**
99% Confidence Level	7	1.33	11	2.09	3	0.57	14	2.66
95% Confidence Level	10	1.90	19	3.61	10	1.90	20	3.80
90% Confidence Level	3	0.57	10	1.90	7	1.33	14	2.66
Insignificant	76	14.42	137	26.00	115	21.82	127	24.10
Total	**96**	18.22	178	33.78	135	25.62	175	33.21

2.3. Methods

This study detected the trends in hydro-meteorological data (i.e., groundwater depth and rainfall) as well as in irrigated areas with the help of MK tests (Mann, 1945; Kendall, 1975) and slope was estimated by Sen's Slope estimator (Sen, 1968). These non-parametric approaches are widely used for detecting trends in hydro-meteorological data such as groundwater (Rahman et al., 2016), rainfall (Tabari et al., 2012; Kamruzzaman et al., 2018; Das and Bhattacharya, 2018; Das et al., 2019, 2020) since it is not necessary to test the distribution of the data and provide robust results than parametric approaches such as linear regression (Tabari et al., 2012; Nalley et al., 2013; Rahman et al., 2018b).

The test statistic (S) of a time series $x_1, x_2, x_3 \ldots$, and x_n can be calculated using MK test as (Mann, 1945; Kendall, 1975):

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^{n} sign(x_j - x_k) \tag{1}$$

Here, n is the quantity of data points, x_j and x_k denotes the data points of time j and k,

$$\begin{aligned} Sign(x_j - x_k) &= +1 \; if \; x_j - x_k > 0 \\ &= 0 \; if \; x_j - x_k = 0 \\ &= -1 \; if \; x_j - x_k < 0 \end{aligned} \tag{2}$$

The variance of test statistics VAR(S) can be estimated as:

$$VAR(S) = \frac{1}{18}\left\{ n(n-1)(2n+5) - \sum_{i=1}^{g} t_i(t_i - 1)(2t_i + 5) \right\} \tag{3}$$

Here, g denotes the tied group's number which is a set of sample data with similar value and t_i indicates the extent ith ties number.

The estimated S and VAR(S) are used to estimate the test statistic Z follows, when n is > 10 (Gilbert, 1987):

$$Z = \begin{cases} \dfrac{S-1}{\sqrt{VAR(S)}} & , if \; S > 0 \\ 0 & , if \; S = 0 \\ \dfrac{S+1}{\sqrt{VAR(S)}} & , if \; S < 0 \end{cases} \tag{4}$$

A positive and negative value of Z specifies the directions of trends such as positive value indicates increasing trend and vice versa.

MK test over estimate or under estimate the Z statistics, if there is a significant autocorrelation at-lag-1 in time series data (Nalley et al., 2013; Rahman et al., 2018b; Kamruzzaman et al., 2016). This study estimated the auto-correlation at lag-1 to confirm about credibility of Z statistics obtained from MK test. If there were significant auto-correlation in the data; this study estimated test statistics (Z) by MMK test (Hamed and Rao, 1998).

The modified VAR (S) statistics can be estimated as (Hamed and Rao, 1998):

$$VAR(S) = \left(\frac{n\,(n-1)(2n+5)}{18}\right) \cdot \left(\frac{n}{n_e^*}\right) \tag{5}$$

Here, the correction factor $\left(\frac{n}{n_e^*}\right)$ is adjusted to the auto-correlated data as:

$$\left(\frac{n}{n_e^*}\right) = 1 + \left(\frac{2}{n^3 - 3n^2 + 2n}\right) \sum_{f=1}^{n-1} (n-f)(n-f-1)(n-f-2)\, \rho_e(f) \tag{6}$$

$\rho_e(f)$ represents the autocorrelation function between ranks of observations and can be estimated as:

$$\rho(f) = 2 \sin\left(\frac{\pi}{6}\, \rho_e(f)\right) \tag{7}$$

Sen's slope (Sen, 1968) estimator was used to estimate the magnitude of change (slope Q). The slope Q can be obtained from Npairs of data as:

$$Q_i = \frac{x_k - x_j}{k - j}, \qquad i = 1, 2, \cdots\cdots N, \quad k > j \tag{8}$$

Where, x_k and x_j represent values of data at k, j times and; Q_i is the median slope respectively.

3. Results and discussions

3.1. Trend in groundwater depth

This study investigated the recent (1996–2017) trends in ground water level depth with the help of MK test/mMK test based on auto-correlation (at lag-1) and Sen's Slope estimator was used for assessing the magnitude change.

3.1.1. Pre-monsoon

Trend test results of pre-monsoon season ground water-level depth time series are presented in Table 1 and Fig. 3a. The obtained results clearly indicated that groundwater depth in the most of the studied wells (n = 527 wells) showed increasing trends (81.78%) and rest of them showed decreasing (18.22%) tendency during the period of 1996–2017. About half (46.87%) of the wells showed significant increasing trends at different confidence intervals (Table 1). Spatial distribution of Z statistics of trend test result is shown in Fig. 3a. The whole study area showed increasing trend in groundwater depth except some pocket areas which are distributed discretely over the study area (Fig. 3a), account of significant decreasing trend was only 3.80% (Table 1). Most of the significant increasing trends were found in southern and central part of the LGRB. Sen's slope analysis revealed that groundwater depth increases

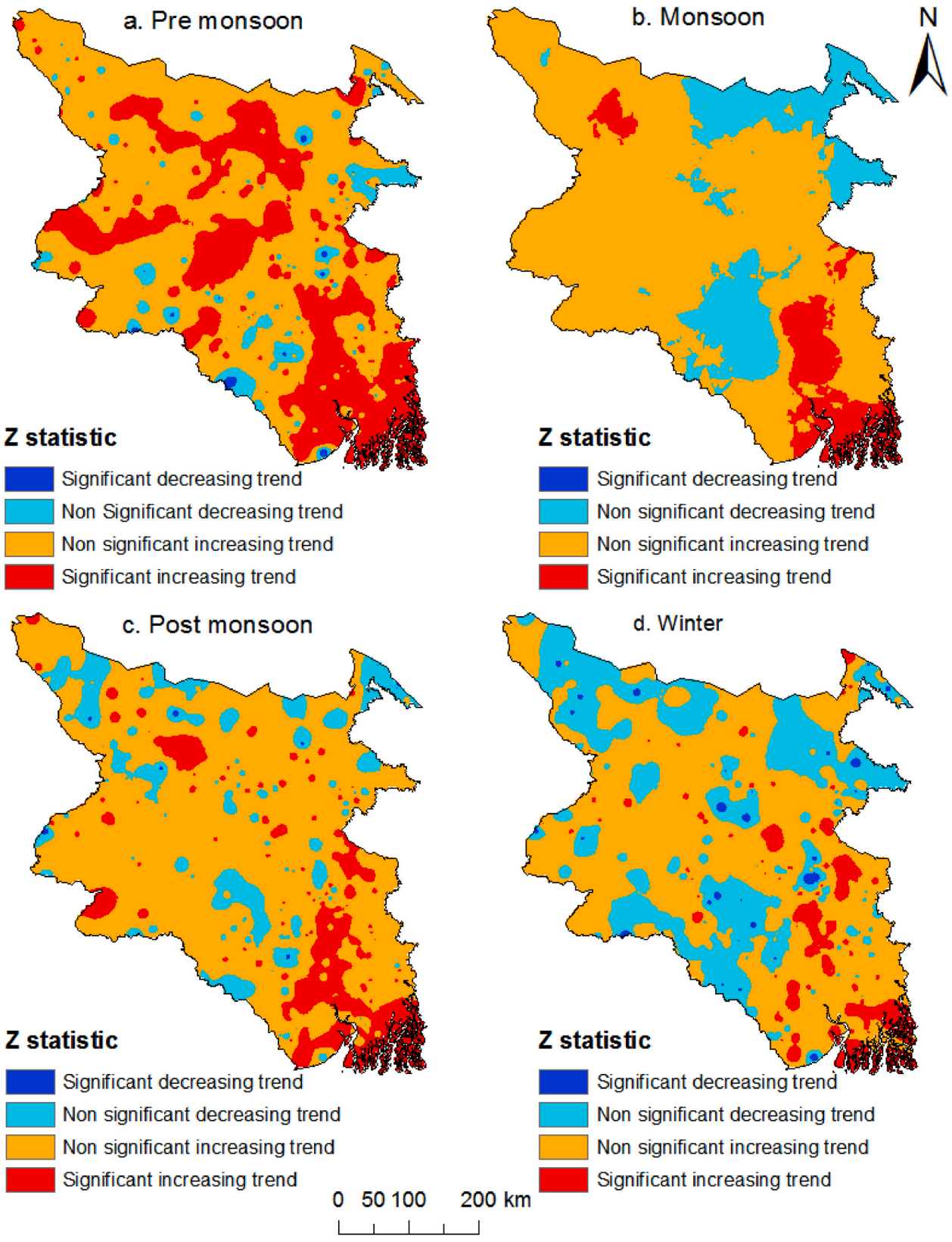


Fig. 3. Distribution of Z statistic of a. Pre-monsoon b. Monsoon c. Post-monsoon and d. winter.

on average at a rate of 10.97 cm/year over the area. Similar to significant trend (Z-statistics) distribution, the higher magnitude of changes (>20 cm/year) were found in the central and southern part of the area (Fig. 5a). The magnitude of increasing trend for the major part of the area varied between 6.83 and 95.32 cm/year. The highest increasing rate (95.32 cm/year) was found for Bamongola dug well located in Malda district in West Bengal and the highest decreasing rate (Sen Q = −70.77 cm/year) was also found in West Bengal at Kutubsahar dug

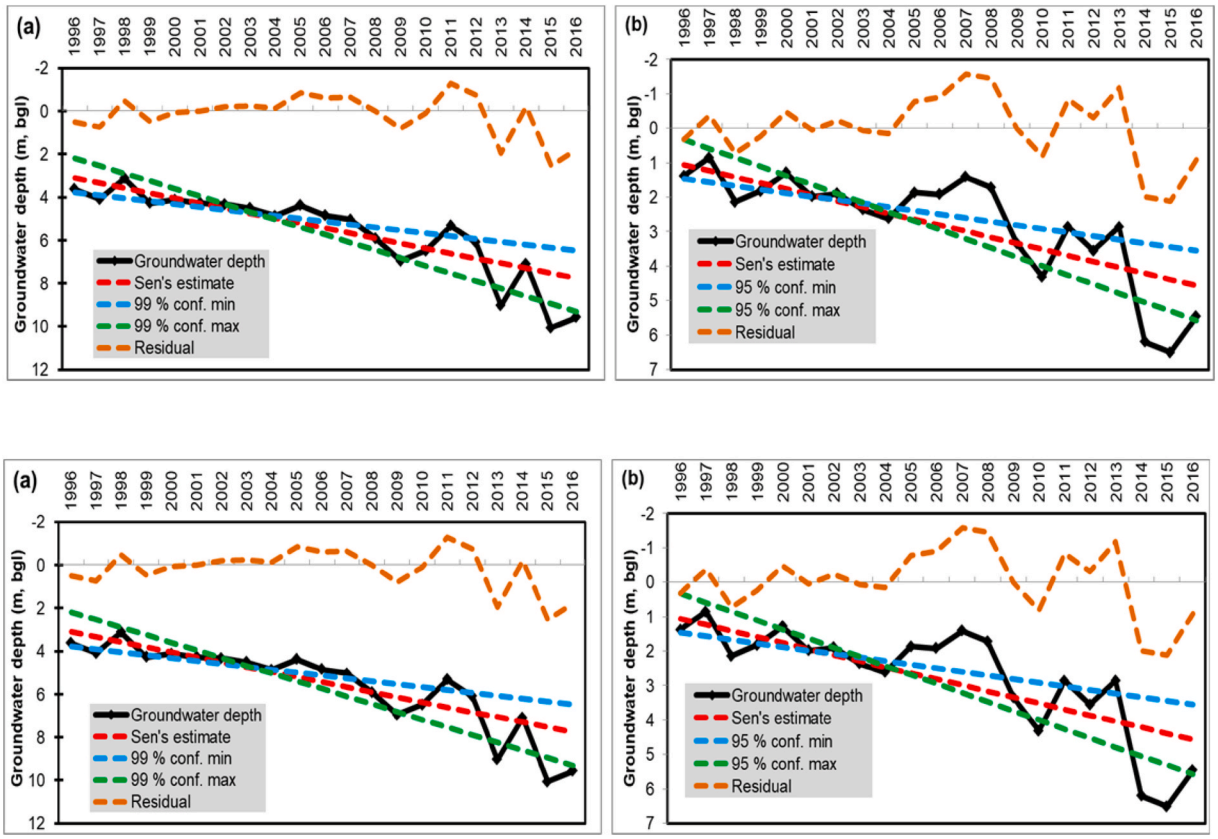


Fig. 4. Rapid increasing trend in groundwater depth in monsoon season (a) dug well in Dhopagachi, and (b) dug well in Rajgir.

well under Malda district. Increasing trends in groundwater depth during pre-monsoon adversely affects the agricultural productions especially *Boro* rice which almost entirely depends on groundwater irrigation. Water requirement during this period is higher due to the higher temperature and evaporation (Tuinenburg et al., 2012). Small farmers rely on shallow pumps for irrigated their lands. Groundwater level may fluctuate below the suction mode of shallow pump in near future due to continuous increasing of groundwater depth and consequently cost of irrigation will increase due to scare shallow groundwater during this high water demand seasons.

3.1.2. Monsoon

The trend analysis results of monsoon ground water depth also showed dominate increasing trends throughout the LGRB during 1996–2017 (Fig. 3b). More than half of the wells (62.22%, n = 527) showed increasing tendency and almost one third of the wells (28.65%) showed a significant increasing trend at different confidence levels (Table 1). The obtained significant increasing trends were mostly found in south-eastern and north-western part of the LGRB. The highest significant increasing trend (104.46 cm/year) of ground water depth was observed at Kuli in Murshidabad district of West Bengal (Fig. 5b). The average rate of increasing of ground water depth obtained from Sen' slope was 7.63 cm/year. Hence, increasing depth of ground water during the rainy season infers that the amount of groundwater recharge of the area is not sufficient for regain the groundwater during this high water season/recharge period. In other words, groundwater abstraction during the dry period is much higher than amount of groundwater recharge during rainy seasons. At the same time, the numbers of observed wells with significant declining trends were very low (7.6%) (Table 1) sparsely distributed over the study area. The highest significant decreasing trend ($-72.02\ cm/year$) was observed in West Bengal at Bishnupur under the district of Bankura. Fig. 4 shows rapid increasing groundwater depth of two most vulnerable observed dug wells of the study area namely Dhopagachi and Rajgir respectively.

3.1.3. Post-monsoon

The spatial distribution of the results of Z statistics trend and Sen's slope of ground water level depths time series of post monsoon are presented in Figs. 3c and 5c, which are more or less similar to the monsoon season. In this period, two third (74.38%, n = 527) of the wells showed positive trends in water-level depth below the ground surface and approximately one third (32.06%) of the wells showed significant positive trend at the confidence level of 99%, 95% and 90% respectively (Table 1). Sen's slope analysis revealed that average groundwater depth increases at the rate of 6.76 cm/year over the area. The degree of increasing trend for the major part of the area varied between 0.01 and 104.44 cm/year. This may due to inadequate recharge of the wells in the monsoon season. However, significant positive trends are sparsely distributed over LGRB with maximum concentrated in the south-eastern, eastern and north-western part (Fig. 3c). On the contrary, a little portion (3.80%) of the observation wells showed significant negative trend during 1996–2017 (Table 1) with the highest significant negative trend (− 64.17 cm/year) for Bishnupur dug wells located in Bankura district of West Bengal.

3.1.4. Winter

The spatiotemporal variation of ground water depth trend and their magnitude in the winter time series are presented in Figs. 3d and 5d. The obtained results clearly revealed that groundwater depth of two-third (66.79%) of the dug wells showed increasing trends and 33.21% rest of the wells showed decreasing trend during 1996–2017 (Table 1). About one fourth (23.91%) of the wells showed significant increasing tendency and only 9.12% wells showed significant decreasing tendency at different confidence levels (Table 1). Most of the significant increasing trends were found in south-eastern, eastern, western and central part of the LGRB and the magnitude of slopes were

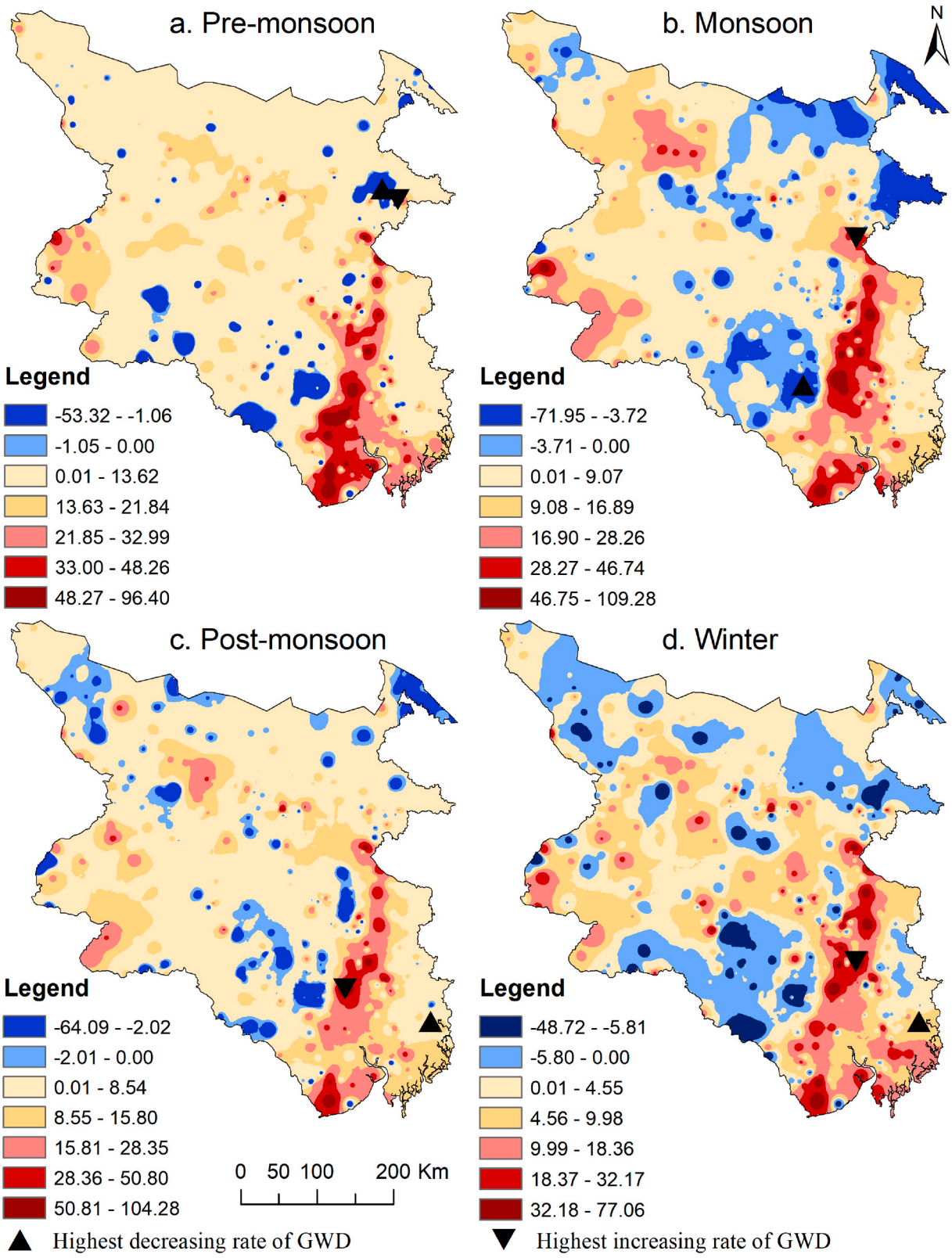


Fig. 5. Distribution of slope of ground water depth with location of maximum of increasing and decreasing dug wells of a. Pre-monsoon b. Monsoon c. Post-monsoon and d. winter.

comparatively lower (0.01–75.78 cm/year) during this season, with an average of 4.30 cm/year for the study area. The highest increasing (75.78 cm/year) trend was observed in West Bengal at Ramjibanpur dug wells under the district of Paschim Midnapore. On the contrary, significant decreasing trends were sparsely distributed in pockets over the LGRB (Fig. 3d) and the magnitude of trend were varied from − 48.83 to − 2.27 cm/year (Fig. 5d).

Table 2
Number of rainfall stations with identified trends by MK & MMK Test in recent (1980–2010) at different confidence level in the study area.

Increasing Trends in rainfall	Pre -monsoon		Monsoon		Post- monsoon		Winter		Annual	
	Number of rainfall stations	% of stations	Number of rainfall stations	% of stations	Number of rainfall stations	% of stations	Number of rainfall stations	% of stations	Number of rainfall stations	% of stations
99% Confidence Level	–	–	1	1.41	4	5.63	–	–	1	1.41
95% Confidence Level	4	5.63	–	–	10	14.08	–	–	–	–
90% Confidence Level	3	4.23	–	–	6	8.45	1	1.41	1	1.41
Insignificant	52	73.24	7	9.86	36	50.70	30	42.25	7	9.86
Total	**59**	83.10	8	11.27	56	78.87	31	43.66	9	12.68
Decreasing Trends in rainfall	**Number of rainfall stations**	**% of stations**	**Number of rainfall stations**	**% of stations**	**Number of rainfall stations**	**% of stations**	**Number of rainfall stations**	**% of stations**	**Number of rainfall stations**	**% of stations**
99% Confidence Level	–	–	28	39.44	–	–	–	–	27	38.03
95% Confidence Level	–	–	18	25.35	1	1.41	1	1.41	11	15.49
90% Confidence Level	–	–	7	9.86	–	–	2	2.82	8	11.27
Insignificant	12	16.90	10	14.08	14	19.72	37	52.11	16	22.54
Total	**12**	16.90	63	88.73	15	21.13	40	56.34	62	87.32

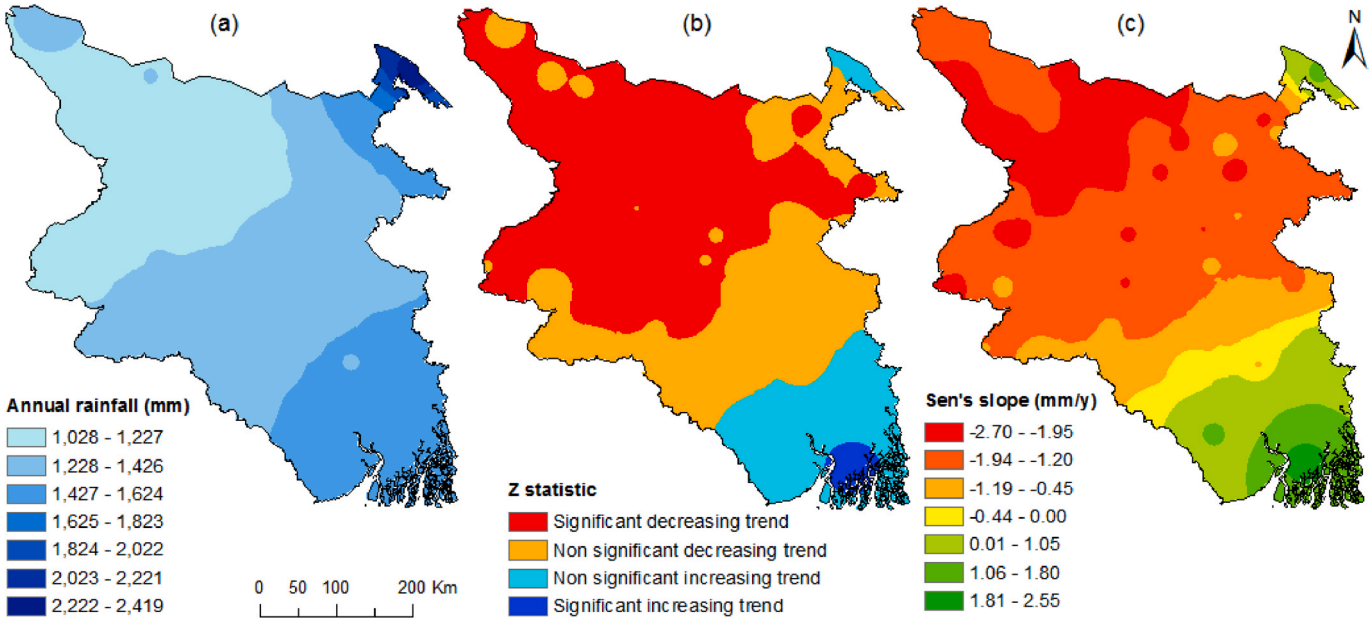


Fig. 6. Distribution of (a) annual rainfall, (b) Z statistic of annual rainfall, and (c) Sen's slope (Q) of annual rainfall.

3.2. Trends in rainfall

3.2.1. Annual trend

The spatial distribution of Z statistics and their magnitude of annual rainfall are presented in Table 2 and Fig. 6b & c. The results clearly indicated that rainfall pattern in most of the stations (total stations n = 71) showed decreasing trends (87.32%) and rest of them showed increasing (12.68%) tendency during the period of 1980–2010. About two third (64.78%) of the stations showed significant decreasing tendency at different confidence level, while the account for significant increasing trend was only 2.82% (Table 2). The spatial distribution of Z-statistics (Fig. 6b) shows that decreasing trends are dominating over the study area except the southern part, and the northern part (up to central part) area shows significant decreasing trends and slope varied from − 2.70 to − 0.44 mm/year. On the other hand, a significant increasing trend was found at South 24 Parganas station during the study period, magnitude of trend was 2.55 mm/year. Bera (2017) also found highest increasing (4.99 mm/year) trend at South 24 Parganas stations.

3.2.2. Seasonal trend

Trend analysis results of seasonal rainfall showed both increasing and decreasing trends throughout the LGRB during 1980–2010. The estimated value of Z statistics were + 0.80 − 1.96, +0.89 and −0.19 for pre-monsoon, monsoon, post-monsoon and winter; respectively. Although the other seasons showed insignificant trend, monsoon season rainfall in the basin area is significant at 95% confidence level. Station wise analysis reveals that the trends of pre-monsoon and post-monsoon are dominated by insignificant increasing trend, accounting for 52% and 50.70%; respectively. Monsoon and winter season were dominated by significant decreasing (74.65%) and insignificant decreasing (52.11%) trends respectively (Table 2). The entire basin area showed an increasing trend in pre-monsoon season rainfall except the extreme

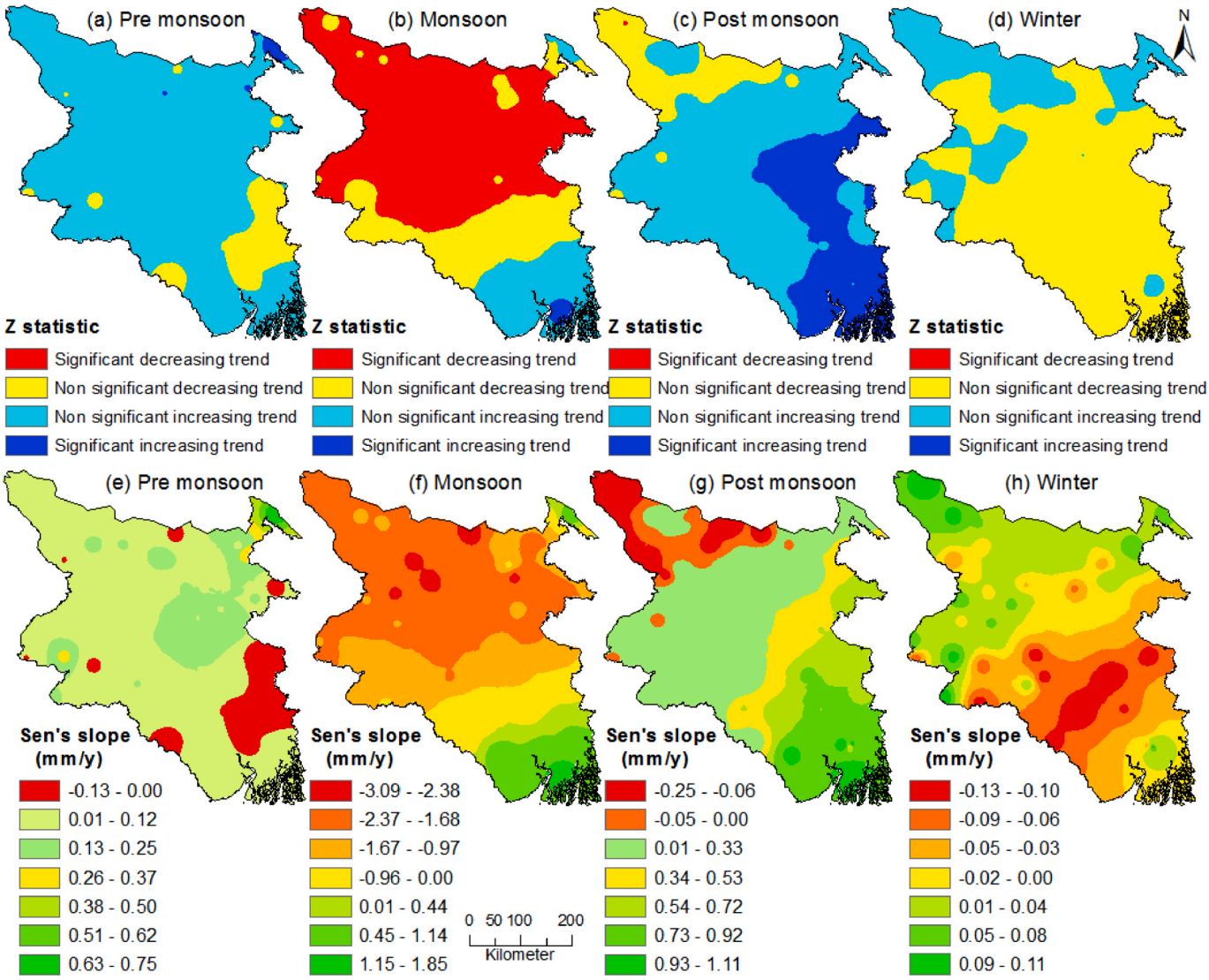


Fig. 7. Distribution of Z statistic and slope (Q, mm/year) of seasonal rainfall.

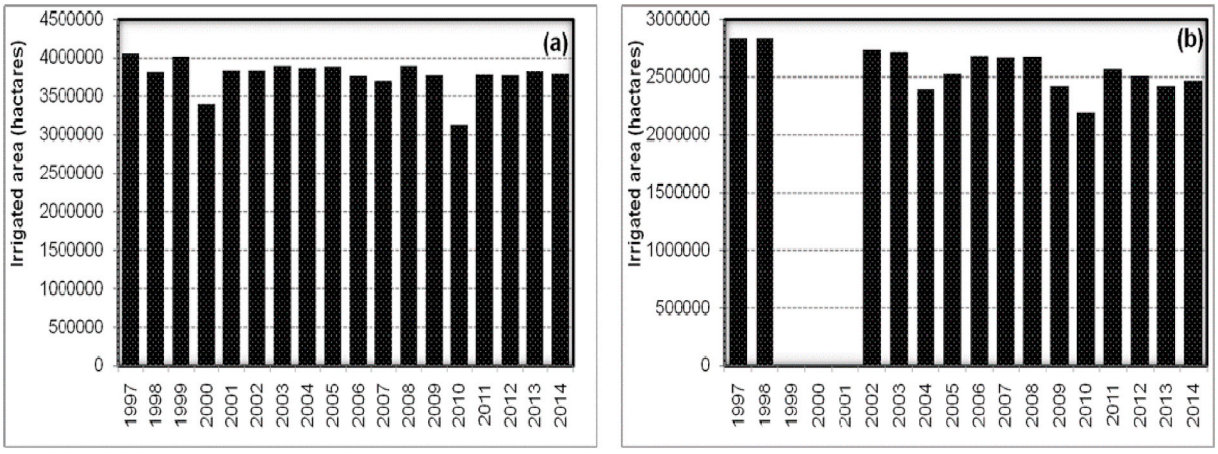


Fig. 8. Time series of *Boro* rice irrigated area (a) West Bengal and (b) Bihar.

south-eastern part (Fig. 7a). Monsoon trend map (Fig. 7b) shows similar spatial distribution of trends for annual rainfall. Insignificant increasing trends in post-monsoon season rainfall were found throughout the basin except the northwestern part (Fig. 7c), whereas insignificant decreasing trend in winter season were observed over whole basin area except few pockets of northern and western part of the basin area (Fig. 7d). Bera

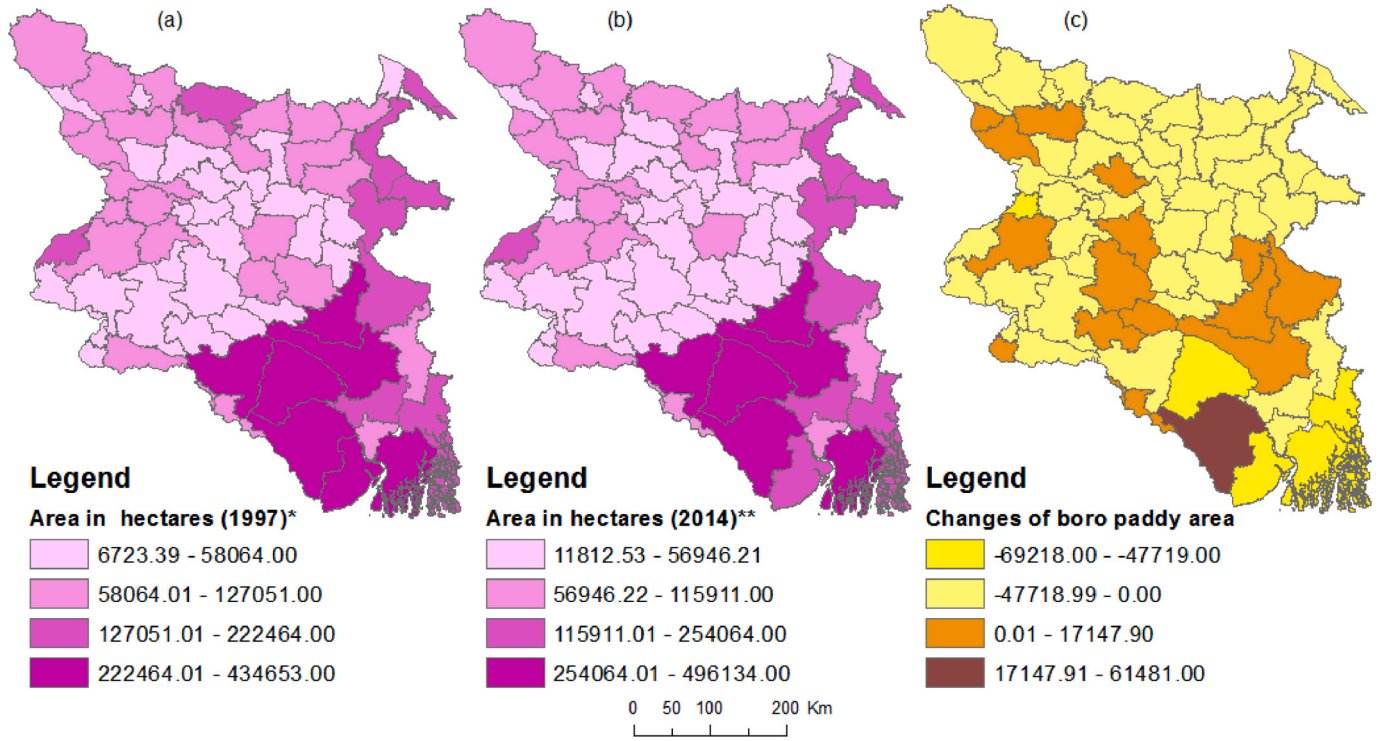


Fig. 9. District wise distribution of *Boro* rice irrigated area of (a) 1997 (For Jharkhand, 2002), (b) 2014 (For Jharkhand, 2007) and (c) Changes between two period.

(2017) also identified significantly decreasing trend in pre-monsoon and monsoon rainfall over Ganga basin during 1901–2000. However, the magnitude of rainfall trend in pre-monsoon, post-monsoon and winter are very low (Fig. 7), while in monsoon season the slope of trend is quite high (1.42 mm/year).

4. Irrigated area

There is no direct record of the amount of annual groundwater abstraction for the study area. This study used the irrigated area to get idea about groundwater abstraction for *Boro* rice irrigation. Fig. 8 shows that there is no increasing trend in *Boro rice* irrigated area during the period 1997–2014. However, it showed a slow decreasing trend both in West Bengal and Bihar state. These results suggest that the trends in *Boro* rice irrigated area cannot alone be used for explanation of groundwater depth increase over the whole area. Our study area also covers another one state which is Jharkhand. However, there is no long-term precise record of the irrigated area like these two states. Fig. 9(a) and (b) show the district wise distribution of irrigated area in LGRB for the years 1997 and 2014; respectively. Fig. 9c shows the changes in irrigated area in LGRB, witnessed the comparatively more positive change at southern (17147.91–61481.00 ha) and less in eastern, central and western (0.01–17147.91 ha) region. Rest of the basin area showed a negative change (−69218.00–0.00 ha) during the study period.

5. Conclusions

This study documented the linkage of groundwater depletion with irrigated area and rainfall under the changing climate in LGRB area. Our study found that shallow groundwater resource in the area is rapidly declining as the depth of the groundwater rapidly increases during rainy season. Furthermore, decreasing trends in rainfall dominated over the area and 64.78% (n = 71) had a significant decreasing trend during the period of 1980–2010. However, the magnitude of change of rainfall was far lower (−2.30 mm/year) than that of increasing rate of groundwater depth. Hence, it can be inferred that groundwater recharge rate is far lower than the amount of groundwater abstraction during dry seasons (winter, pre-monsoon) and decreasing trend in rainfall fasten the groundwater depletion. For example, the rapid increasing groundwater level depth in winter and pre-monsoon indicates more pumping during these seasons, causing a larger number of observation wells to have a declining trend in these seasons. As there are no significant trends in irrigated areas, it can also infer that groundwater development occurs without considering proper hydrological planning-water balance *i.e.* recharge versus abstraction. The cultivation of *Boro* paddy which is completely dependent upon the irrigation will face difficulties in near future. Hence, advance irrigation water management techniques such as raised bed, alternate drying and wetting need to promote for saving water as well as surface water irrigation area need to increase to minimize the stress on groundwater resource. To achieve sustainability in groundwater resource, water resource managers need to prepare and execute integrated water resources management plan and new norms for the abstraction of groundwater for the area.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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